

2012



Mathematics

Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen.
- Write your student number on each booklet.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- The mark allocated for each question is listed at the side of the question.
- Marks may be deducted for careless or poorly presented work.

Total Marks – 100

Section I – Pages 2 – 4

10 marks

- Attempt Questions 1 – 10.
- Allow about 15 minutes for this section.
- Answer on the sheet provided.

Section II – Pages 5 – 13

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section.
- Begin each question in a new booklet.
- Show all necessary working in Questions 11 – 16.
- Templates for Q12(a) to be detached and placed in answer booklet.

Q12.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Section I – (10 marks)

Marks

Answer this section on the answer sheet provided at the back of this paper.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

1. The maximum value of y reached by the ellipse with equation

$$\frac{3(x+3)^2}{5} + \frac{(y-4)^2}{6} = 3$$

is:

- A. $-4 + 3\sqrt{2}$
B. $4 + \sqrt{5}$
C. $3\sqrt{2}$
D. $4 + 3\sqrt{2}$
2. The graph of $f(x) = \frac{1}{x^2 + mx - n}$, where m and n are real constants, has no vertical asymptotes if
- A. $m^2 < 4n$
B. $m^2 > 4n$
C. $m^2 = -4n$
D. $m^2 < -4n$
3. The number of real solutions to $x^4 - x^3 = \operatorname{cosec}^2(x) - \cot^2(x)$ is:
- A. 0
B. 1
C. 2
D. 3
4. If $z = \frac{3+4i}{1+2i}$, the imaginary part of z is:
- A. -2 B. $-\frac{2}{5}i$ C. $-\frac{2}{5}$ D. $-2i$

Section I (cont'd)

Marks

5. If $I = \int_0^{\ln 2} \frac{e^x}{e^x + e^{-x}} dx$ and $J = \int_0^{\ln 2} \frac{e^{-x}}{e^x + e^{-x}} dx$, then the exact value of $I - J$ is:
- A. $\ln\left(\frac{5}{2}\right)$ B. $\ln 2$ C. $\ln(5)$ D. $\ln\left(\frac{5}{4}\right)$
6. If $z = \sqrt{3} + i$ then in modulus/argument form $z = 2\text{cis}\frac{\pi}{6}$. If $z^n + (\bar{z})^n$ is to be rational, then the integer 'n' can not be:
- A. 2
B. 3
C. 5
D. 6
7. Given hyperbola $\mathcal{H}$ with equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ has eccentricity e then the ellipse E with equation $\frac{x^2}{a^2+b^2} + \frac{y^2}{b^2} = 1$ has eccentricity.
- A. $-e$ B. $\frac{1}{e}$ C. $\sqrt{e}$ D. e^2
8. What restrictions must be placed on p if α, β, γ are the three, non-zero real roots of the equation $x^3 + px - 1 = 0$?
- A. $p > 0$, p is real
B. $p < 0$, p is real
C. $p \geq 0$, p is real
D. $p \leq 0$, p is real

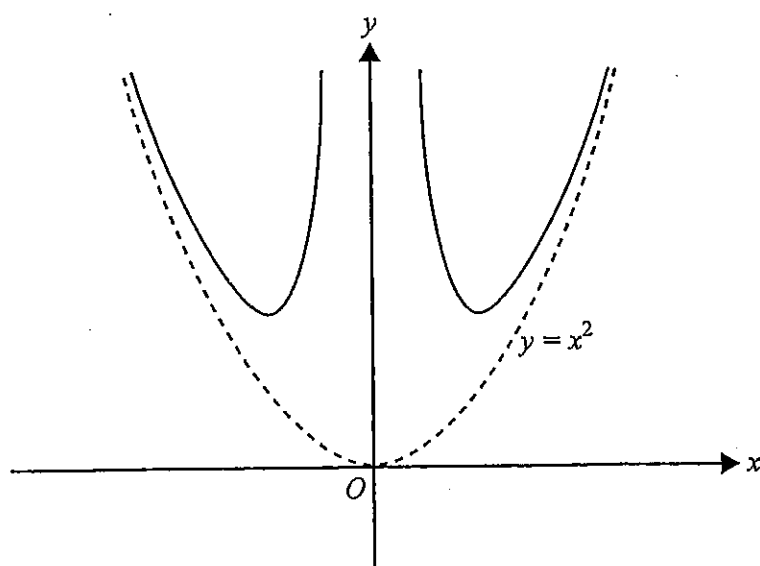
Section I (cont'd)

Marks

9. Given that $\frac{dy}{dx} = y^2 + 1$, and that $y = 1$ at $x = 0$, then

- A. $y = \tan\left(x - \frac{\pi}{4}\right)$
- B. $y = \tan\left(x + \frac{\pi}{4}\right)$
- C. $x = \log_e\left(\frac{y^2+1}{2}\right)$
- D. $y = \frac{1}{3}y^3 + y - \frac{1}{3}$

10.



A possible equation for the graph of the curve shown above is

- A. $y = \frac{x^3+a}{x}, \quad a > 0$
- B. $y = \frac{x^3+a}{x}, \quad a < 0$
- C. $y = \frac{2x^4+a}{x^2}, \quad a > 0$
- D. $y = \frac{x^4+a}{x^2}, \quad a < 0$

Section II – Show all working

Question 11 – Start A New Booklet – (15 marks)

Marks

a) Find $\int \frac{dx}{\sqrt{3-4x-4x^2}}$ 2

b) Evaluate $\int_0^{\frac{\pi}{6}} \frac{d\theta}{9-8\cos^2\theta}$ using the substitution $t = \tan \theta$ 3

c) Find $\int \frac{dx}{(x+1)(x^2+4)}$ 3

d) Evaluate $\int_0^1 \tan^{-1}x \, dx$ 2

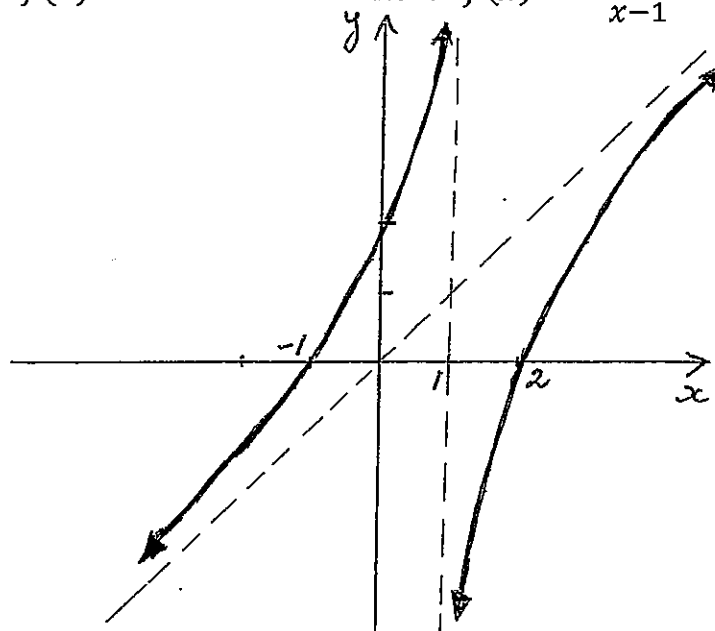
e) If $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$ show that $I_n = \frac{n-1}{n} \cdot I_{n-2}$ 3

Hence evaluate $\int_0^{\frac{\pi}{2}} \sin^5 x \, dx$ 2

Question 12 – Start A New Booklet – (15 marks)

Marks

- a) The sketch of $y = f(x)$ is shown below where $f(x) = \frac{x^2 - x - 2}{x - 1}$



- (i) Show that $y = x$ is an asymptote. 2
- (ii) Sketch each of the following on the template provided.
- (α) $y = |f(x)|$ 2
- (β) $y = f(1 - x)$ 2
- (γ) $y^2 = f(x)$ 2
- b) Consider the curve $C: x^2 + xy + y^2 = 9$
- (i) Find $\frac{dy}{dx}$ 1
- (ii) Find all stationary points and points where $\frac{dy}{dx}$ is not defined. 4
- (iii) Sketch C clearly showing the above features and intercepts on the x, y axes. 2

Question 13 – Start A New Booklet – (15 marks)

Marks

a) If $z = (1 + i)^{-1}$.

(i) Express $\bar{z}$ in modulus-argument form.

2

(ii) If $(\bar{z})^9 = a + ib$ where a and b are real numbers, find the values of a and b

2

b) Sketch each of the following on separate Argand diagrams.

(i) $|z - 2 + 3i| = |z + 2 - 3i|$

2

(ii) $\arg(z + 3 - i) = \frac{3\pi}{4}$

2

c) (i) On an Argand diagram sketch $|z - \sqrt{2} - \sqrt{2}i| = 1$

2

(ii) Find the minimum values of $|z|$ and $\arg z$

3

d)

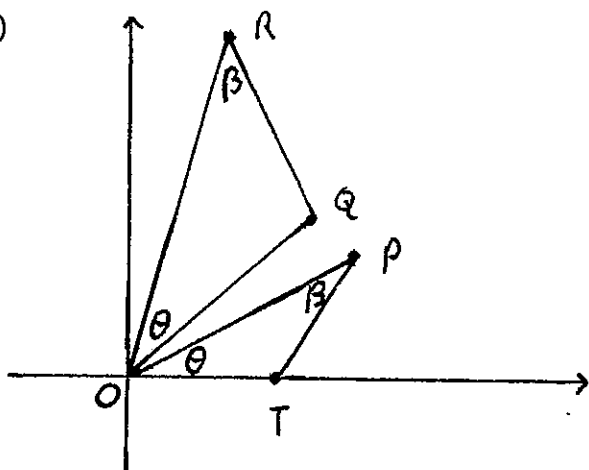


Fig I

The points T, P and Q in the complex plane correspond to the complex numbers $1, \sqrt{3} + i$ and $2 + 2i$ respectively.

2

Triangles OTP and OQR are similar with corresponding angles as shown in Fig I. Find the complex number represented by R (in modulus argument form).

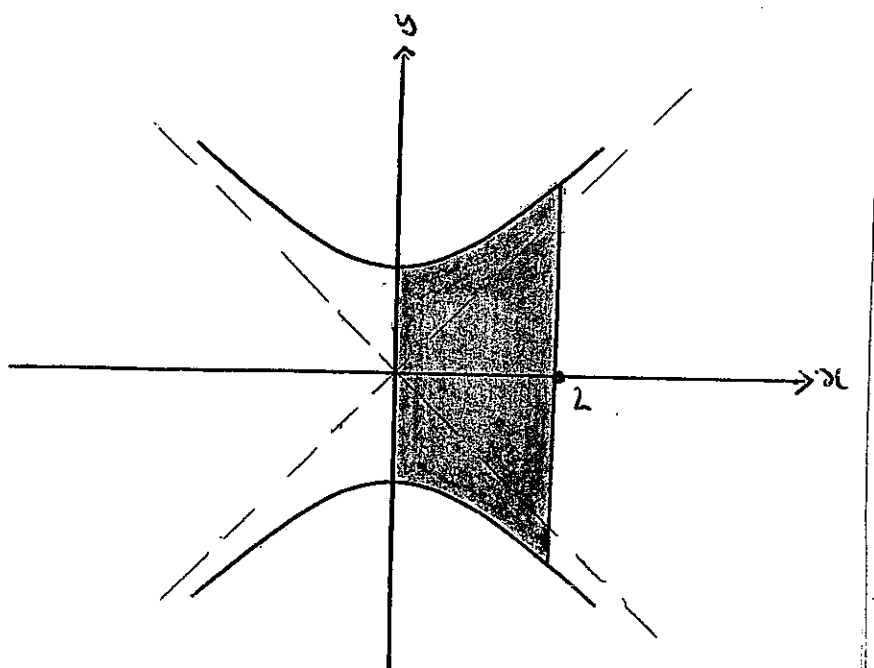
Question 14 – Start A New Booklet – (15 marks)		Marks
a)	The polynomial equation $x^3 - 6x^2 + 3x - 2 = 0$ has roots α, β, γ . Evaluate $\alpha^3 + \beta^3 + \gamma^3$	2
b)	Prove that if a polynomial $P(x)$ has a zero of multiplicity ' m ' then the derived polynomial $P'(x)$ has that same zero with multiplicity ' $m - 1$ '	1
c)	Given that $-2 - i$ is a zero of $P(x) = x^4 + 6x^3 + 14x^2 + 14x + 5$, find all zeros of $P(x)$	3
d)	(i) Prove that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ by use of de Moivre's theorem.	2
	(ii) Find the general solution of $\cos 3\theta = \frac{1}{2}$	1
	(iii) Solve for $x : 8x^3 - 6x - 1 = 0$	3
	(iv) Find a polynomial of least degree which has zeros $\sec^2 \frac{\pi}{9}, \sec^2 \frac{5\pi}{9}, \sec^2 \frac{7\pi}{9}$	2
	(v) Hence evaluate $\sec^2 \frac{\pi}{9} + \sec^2 \frac{5\pi}{9} + \sec^2 \frac{7\pi}{9}$	1

Question 15 – Start A New Booklet – (15 marks)

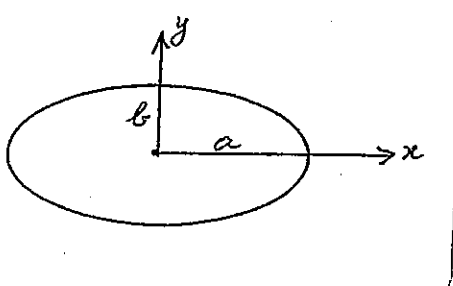
Marks

- a) Using the method of cylindrical shells, find the volume generated by revolving the area bounded by the lines $\begin{cases} x = 2 \\ x = 0 \end{cases}$ and the two branches of the hyperbola $\frac{y^2}{9} - \frac{x^2}{4} = 1$ about the y -axis (as shown in the diagram)

3



- b) (i)



The ellipse shown has equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

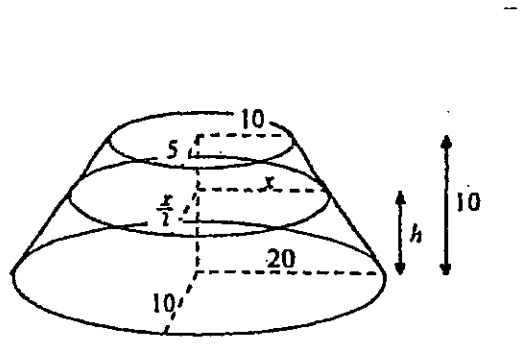
Prove that the area enclosed by this ellipse is πab

3

Question 15 (cont'd)

Marks

b) (ii)



A solid of height 10 m stands on horizontal ground.

- The base of the solid is an ellipse with semi-axes of 20 m and 10 m.
- The top of the solid is an ellipse with semi-axes of 10 m and 5 m.

Horizontal cross-sections taken parallel to the base and at height h metres above the base are ellipses with semi-axes x metres and $\frac{x}{2}$ metres.

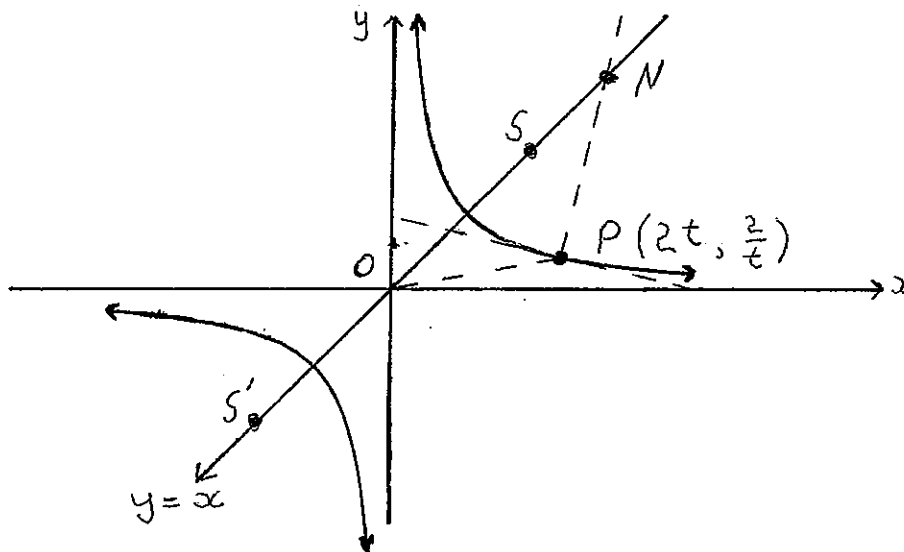
The centres of these elliptical cross-sections and the base lie on a vertical straight line, and the extremities of their semi-axes lie on sloping straight lines as shown in the diagram.

- (α) Prove that $x = 20 - h$ 2
- (β) Find the volume of the solid correct to the nearest cubic metre. 3

Question 15 (cont'd)

Marks

- c) The diagram shows the hyperbola $xy = 4$



- (i) What are the coordinates of the foci S and S' ?

1

- (ii) The point $P(2t, \frac{2}{t})$ lies on the curve, where $t \neq 0$. The normal at P intersects the straight line $y = x$ at N . O is the origin.

Given the equation of the normal at P is $y = t^2x + \frac{2}{t} - 8$

- (α) Find the coordinates of N

1

- (β) Show that the triangle OPN is isosceles

2

Question 16 – Start A New Booklet – (15 marks)

Marks

- a) A parachutist of mass M is initially located travelling downward in a straight line with a speed of v_0 . [let $x = 0$ at $t = 0$]

If the resistance on the parachute is proportional to the speed and the gravitational force is g .

- (i) Show that the speed, v , can be given as

$$v = \frac{g}{k} - \left(\frac{g}{k} - v_0 \right) e^{-kt}$$

3

(k) is constant of proportionality.

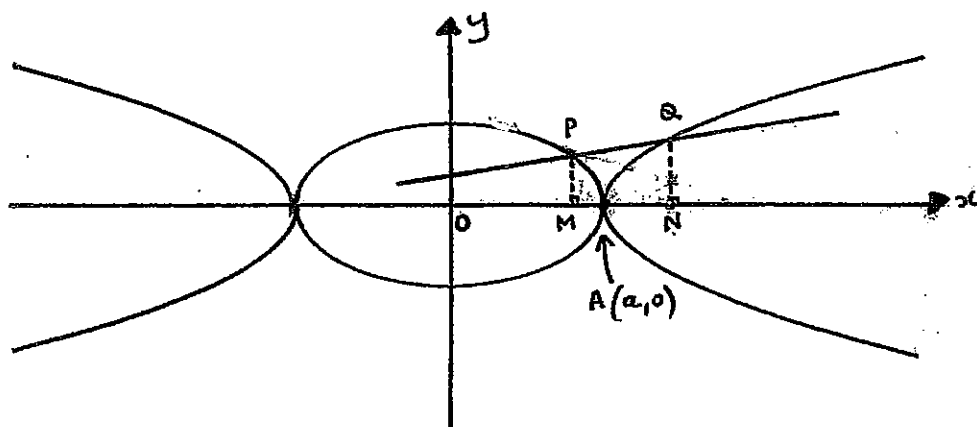
- (ii) Find the parachutist's "terminal" velocity.

1

Questions 16 b) continued on next page

Question 16 (cont'd)

- b) $P(a\cos\theta, b\sin\theta)$ and $Q(a\sec\theta, b\tan\theta)$ lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, respectively as shown.



M and N are the feet of the perpendicular from P and Q respectively to the x -axis. $0 < \theta < \frac{\pi}{2}$, and QP meets the x -axis at K . A is the point $(a, 0)$.

- (i) Given $\triangle KPM \parallel \triangle KQN$, show that $\frac{KM}{KN} = \cos\theta$ 1

- (ii) Hence, show that K has coordinates $(-a, 0)$ 2

- (iii) Show that the tangent to the ellipse at P has equation $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$, and deduce it passes through N 3

- (iv) Given that the tangent to the hyperbola at Q has equation $\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$, show that the tangent passes through M . 2

If T is the point of intersection of PN and QM , show that AT is perpendicular to the x -axis.

- c) Using mathematical induction prove that 3

$$\sum_{r=1}^n r^3 < n^2(n+1)^2$$

Student Number: _____ Teacher: _____

Student Name : _____

Year 12 Mathematics Extension 2 Trial HSC Examination 2012

Section I

Multiple-choice Answer Sheet – Questions 1 – 10

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C ☐ D ☐
 correct

- | | | | | |
|-----|-------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1. | A ☐ | B ☐ | C ☐ | D ☒ |
| 2. | A ☐ | B ☐ | C ☐ | D ☒ |
| 3. | A ☐ | B ☐ | C ☒ | D ☐ |
| 4. | A ☐ | B ☐ | C ☒ | D ☐ |
| 5. | A ☐ | B ☐ | C ☐ | D ☒ |
| 6. | A ☐ | B ☐ | C ☒ | D ☐ |
| 7. | A ☐ | B ☒ | C ☐ | D ☐ |
| 8. | A ☐ | B ☒ | C ☐ | D ☐ |
| 9. | A ☐ | B ☒ | C ☐ | D ☐ |
| 10. | A ☐ | B ☐ | C ☒ | D ☐ |

Trial HSC EXT 2 - 2012

SECTION I

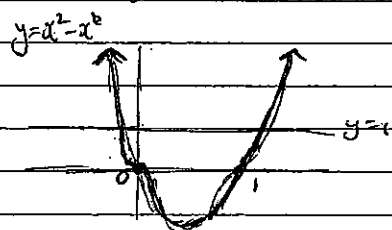
1. $\frac{(x+3)^2}{15} + \frac{(y-4)^2}{18} = 1$ Ellipse centre $(-3, 4)$
 $a = \sqrt{15}$
 $b = \sqrt{18}$
 $= 3\sqrt{2}$

$\therefore$ maximum value of $y = 4 + 3\sqrt{2}$

2. No vertical asymptotes then $x^2 + mx - n \neq 0$

Then $\Delta < 0$, so $m^2 - 4 \cdot 1 \cdot -n < 0$
 $m^2 < -4n$

3. $x^3(x-1) = 1$
 $x^4 - x^3 - 1 = 0$



sketch $y = x^4 - x^3$
 two pts of intersection
 or $y = x^4 - x^3 - 1$
 $y' = 4x^3 - 3x^2$
 $= x^2(4x - 3)$
 $x' = 12x^2 - 6x$
 $= 6x(2x - 1)$
 etc.

4. $z = \frac{3+4i}{1+2i} \times \frac{1-2i}{1-2i}$
 $= \frac{3-6i+4i+8}{1+4}$
 $= \frac{11-2i}{5}$

imaginary part $\frac{2}{5}$

5. $I - J = \int_0^{\ln 2} \frac{e^x - e^{-x}}{e^x + e^{-x}} dx \quad \left[\frac{\ln(x)}{x} \right]$

$= \left[\ln(e^x + e^{-x}) \right]_0^{\ln 2}$
 $= \ln(e^{\ln 2} + e^{-\ln 2}) - \ln(1+1)$
 $= \ln\left[2 + \frac{1}{2}\right] - \ln(2)$
 $= \ln\left(\frac{5}{4}\right)$

6. $z^n + (\bar{z})^n = 2\left(\cos\left(\frac{n\pi}{6}\right) + i\sin\left(\frac{n\pi}{6}\right)\right) + 2\left(\cos\left(\frac{n\pi}{6}\right) - i\sin\left(\frac{n\pi}{6}\right)\right)$
 $= 4\cos\left(\frac{n\pi}{6}\right)$

$n=2, 4\cos\frac{\pi}{3} = \frac{1}{2} \times 4 = 2$

$n=3, 4\cos\frac{\pi}{2} = 0 \times 4 = 0$

$n=5, 4\cos\frac{5\pi}{6} = -\frac{\sqrt{3}}{2} \times 4 = -2\sqrt{3}$ ✓

$n=6, 4\cos\pi = -1 \times 4 = -4$

7. $b^2 = a^2(e^2 - 1)$

$\therefore e^2 = 1 + \frac{b^2}{a^2}$
 $= \frac{a^2 + b^2}{a^2}$

then for E

$b^2 = (a^2 + b^2)(1 - E^2)$

$\therefore E = \frac{1}{e}$

$\frac{b^2}{a^2 + b^2} = 1 - E^2$

$E^2 = 1 - \frac{b^2}{a^2 + b^2}$

$E^2 = 1 - \frac{b^2}{e^2 a^2}$

$= 1 - \frac{1}{e^2} (e^2 - 1)$

$= 1 - 1 + \frac{1}{e^2}$

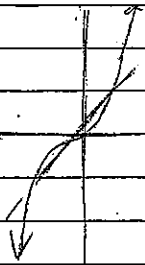
$= \frac{1}{e^2}$

QUESTION 11:

8. Required curve $x^3 + px - 1 = 0$
to cut x-axis at three distinct
places.

Consider
 $x^3 = 1 - px$ [p gradient of line]
 $p < 0$

B



9. $\frac{dx}{dy} = \frac{1}{y^2 + 1}$

$x = \tan^{-1} y + C$

B

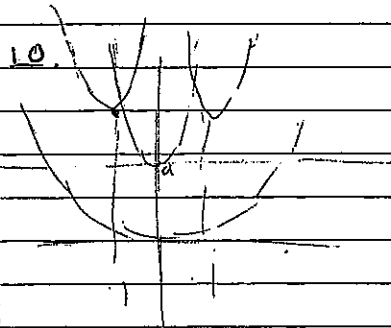
at $x=0, y=1$

$0 = \frac{\pi}{4} + C$

$C = -\frac{\pi}{4}$

$\therefore x + \frac{\pi}{4} = \tan^{-1} y$

$y = \tan(x + \frac{\pi}{4})$



C.

(a) $\int \frac{dx}{\sqrt{3-4x-4x^2}} = \int \frac{dx}{\sqrt{-1(4x^2+4x-3)}}$

$= \int \frac{dx}{\sqrt{-1[(2x+1)^2-4]}}$

* $\int \frac{dx}{\sqrt{2^2-x^2}} \quad \begin{matrix} u=2x+1 \\ du=2dx \end{matrix}$

$= \frac{1}{2} \int \frac{du}{\sqrt{2^2-u^2}}$

$= \frac{1}{2} \sin^{-1}(\frac{u}{2}) + C$

$= \frac{1}{2} \sin^{-1}(\frac{2x+1}{2}) + C$

$= \int \frac{dx}{\sqrt{4-(2x+1)^2}}$

let $2x+1 = 2 \sin \theta$
 $2dx = 2 \cos \theta d\theta$
 $dx = \cos \theta d\theta$

$= \int \frac{\cos \theta d\theta}{2 \cos \theta}$

$= \frac{\theta}{2} + C$

$= \frac{1}{2} \sin^{-1}(\frac{2x+1}{2}) + C$

(b) $\int_0^{\frac{\pi}{6}} \frac{d\theta}{9-8\cos^2 \theta}$

$= \int_0^{\frac{\pi}{6}} \frac{\frac{dt}{1+t^2}}{9-8(\frac{1}{1+t^2})}$

$= \int_0^{\frac{1}{\sqrt{3}}} \frac{dt}{9+9t^2-8}$

* $\int_0^{\frac{1}{\sqrt{3}}} \frac{dt}{1+(3t)^2} \quad \begin{matrix} 3t = \tan \alpha \\ 3dt = \sec^2 \alpha d\alpha \end{matrix}$

$= \int_0^{\frac{\pi}{3}} \frac{1}{3} \cdot \frac{\sec^2 \alpha d\alpha}{\sec^2 \alpha}$

$= \frac{1}{3} \cdot \left[\alpha \right]_0^{\frac{\pi}{3}}$

$= \frac{\pi}{9}$

$t = \tan \theta$ ——— ①
 $dt = \sec^2 \theta d\theta$

ie $d\theta = \frac{dt}{\sec^2 \theta}$
 $= \frac{dt}{1+t^2}$

from ①: $\sec^2 \theta = 1+t^2$
 $\therefore \cos^2 \theta = \frac{1}{1+t^2}$

* $\frac{1}{9} \int_0^{\frac{1}{\sqrt{3}}} \frac{dt}{(\frac{1}{3})^2 + t^2}$
 $= \frac{1}{9} \left[\frac{1}{\frac{1}{3}} \tan^{-1} \left(\frac{t}{\frac{1}{3}} \right) \right]_0^{\frac{1}{\sqrt{3}}}$
 $= \frac{1}{3} \left[\tan^{-1} \left(\frac{3}{\sqrt{3}} \right) - \tan^{-1} 0 \right]$
 $= \frac{1}{3} \times \frac{\pi}{3} = \frac{\pi}{9}$

$$(c) \int \frac{dx}{(x+1)(x^2+4)}$$

$$\begin{aligned} \text{let } \frac{1}{(x+1)(x^2+4)} &= \frac{a}{x+1} + \frac{bx+c}{x^2+4} \\ \text{ie } 1 &= a(x^2+4) + (bx+c)(x+1) \\ x=-1 \Rightarrow 1 &= 5a \\ \therefore a &= \frac{1}{5} \\ \text{co-eff of } x^2 \Rightarrow 0 &= a+b \\ \therefore b &= -\frac{1}{5} \\ \text{constant} \Rightarrow 1 &= 4a+c \\ &= \frac{4}{5} + c \\ \therefore c &= \frac{1}{5} \end{aligned}$$

$$= \int \left(\frac{\frac{1}{5}}{x+1} + \frac{\frac{1}{5} - \frac{1}{5}x}{x^2+4} \right) dx$$

$$= \frac{1}{5} \ln|x+1| - \frac{1}{5} \int \frac{x-1}{x^2+4} dx$$

$$= \frac{1}{5} \ln|x+1| - \frac{1}{5} \int \left(\frac{1}{2} \cdot \frac{2x}{x^2+4} - \frac{1}{x^2+4} \right) dx$$

$$= \frac{1}{5} \ln|x+1| - \frac{1}{10} \ln|x^2+4| + \frac{1}{5} \cdot \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + c$$

$$= \frac{1}{5} \ln|x+1| - \frac{1}{10} \ln(x^2+4) + \frac{1}{10} \tan^{-1}\left(\frac{x}{2}\right) + c$$

$$\begin{aligned} (d) \int_0^1 \frac{1 \cdot \tan^{-1}x}{\frac{du}{dx} \cdot v} dx &= x \tan^{-1}x \Big|_0^1 - \int_0^1 x \cdot \frac{1}{1+x^2} dx \\ &= (1 \tan^{-1}1 - 0) - \frac{1}{2} \left[\ln(1+x^2) \right]_0^1 \\ &= \frac{\pi}{4} - \left(\frac{1}{2} \ln 2 - \frac{1}{2} \ln 1 \right) \\ &= \frac{\pi}{4} - \frac{1}{2} \ln 2 \end{aligned}$$

$$(d) I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$$

$$= \int_0^{\frac{\pi}{2}} \underbrace{\sin x}_{du} \cdot \underbrace{\sin^{n-1} x}_v dx$$

$$= \left[\cos x \cdot \sin^{n-1} x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} -\cos x \cdot (n-1) \sin x \cdot \cos x dx$$

$$= 0 + \int_0^{\frac{\pi}{2}} (n-1) \cdot \cos^2 x \cdot \sin^{n-2} x dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} (1 - \sin^2 x) \sin^{n-2} x dx$$

$$= (n-1) \cdot I_{n-2} - (n-1) I_n$$

$$\text{ie } I_n + (n-1) I_n = (n-1) I_{n-2}$$

$$I_n (1+n-1) = (n-1) I_{n-2}$$

$$\therefore I_n = \left(\frac{n-1}{n} \right) \cdot I_{n-2}$$

$$\text{Then } \int_0^{\frac{\pi}{2}} \sin^5 x dx = I_5$$

$$= \frac{4}{5} \times I_3$$

$$= \frac{4}{5} \times \frac{2}{3} \times I_1$$

$$= \frac{8}{15} \int_0^{\frac{\pi}{2}} \sin x dx$$

$$= \frac{8}{15} \cdot [-\cos x]_0^{\frac{\pi}{2}}$$

$$= \frac{8}{15} [0 - -1]$$

$$= \frac{8}{15}$$

QUESTION 12:

④ (i) $y = f(x) = \frac{(x-2)(x+1)}{(x-1)}$

$$= \frac{x^2 - x}{x-1} - \frac{2}{x-1}$$

$$= \frac{x(x-1)}{x-1} - \frac{2}{x-1}$$

$$= x - \frac{2}{x-1} \quad x \neq 1$$

as $x \rightarrow \pm \infty$, $\frac{2}{x-1} \rightarrow 0$

$\therefore y = x$ is an asymptote

* See template (ii)

⑤ (i) $2x + 1 \cdot y + x \cdot \frac{dy}{dx} + 2y \frac{dy}{dx} = 0$

then $\frac{dy}{dx} (x+2y) = -(2x+y)$

$$\frac{dy}{dx} = -\frac{(2x+y)}{(x+2y)}$$

(ii) Stationary when $2x+y=0$
i.e. $y = -2x$

Then in C: $x^2 + x(-2x) + (-2x)^2 = 9$
 $x^2 - 2x^2 + 4x^2 = 9$

$$3(x^2 - 3) = 0$$

$x = -\sqrt{3}$ and $x = \sqrt{3}$

$y = 2\sqrt{3}$

$y = -2\sqrt{3}$

Stationary $(-\sqrt{3}, 2\sqrt{3})$

$(\sqrt{3}, -2\sqrt{3})$

Not defined when $x+2y=0$
 $x = -2y$

Sub in C: $4y^2 - 2y^2 + y^2 = 9$

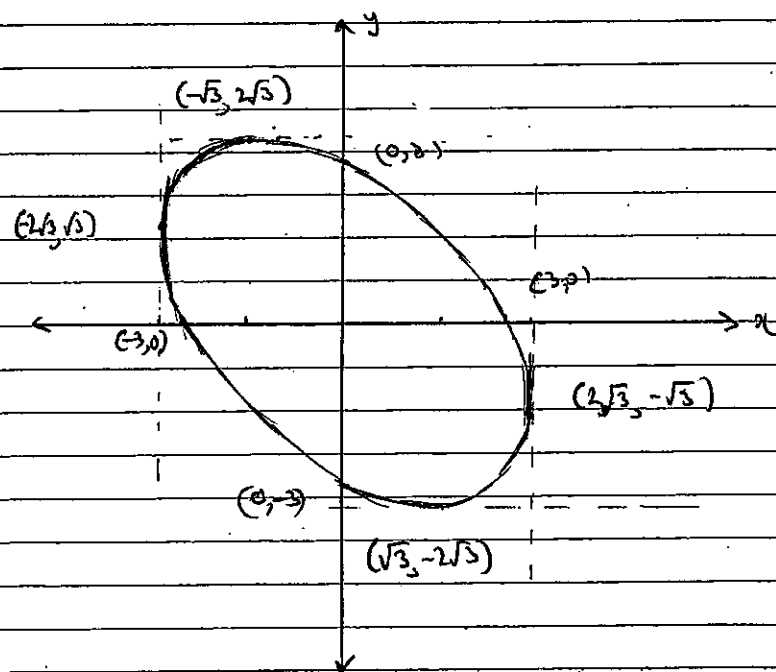
$$3(y^2 - 3) = 0$$

when $y = -\sqrt{3}$
 $x = 2\sqrt{3}$

or $y = \sqrt{3}$
 $x = -2\sqrt{3}$

Not defined at $(2\sqrt{3}, -\sqrt{3})$ and $(-2\sqrt{3}, \sqrt{3})$

(iii)



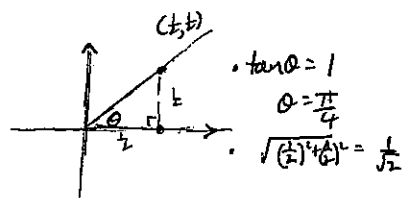
Intercepts: when $x=0$, $y^2=9$ $(0, 3)$; $(0, -3)$

$y=0$; $x^2=9$ $(-3, 0)$; $(3, 0)$

QUESTION 1.3:

$$(a) \quad z = \frac{1}{1+i} \cdot \frac{1-i}{1-i}$$

$$= \frac{1}{2} - \frac{1}{2}i$$



$$(i) \quad \bar{z} = \frac{1}{2} + \frac{1}{2}i$$

$$= \frac{1}{\sqrt{2}} \cos \frac{\pi}{4} \quad \left[\text{ie } \frac{1}{\sqrt{2}} (\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}) \right]$$

$$(ii) \quad (\bar{z})^9 = \left(\frac{1}{\sqrt{2}} \right)^9 \cos \frac{9\pi}{4}$$

$$= \frac{1}{16\sqrt{2}} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right)$$

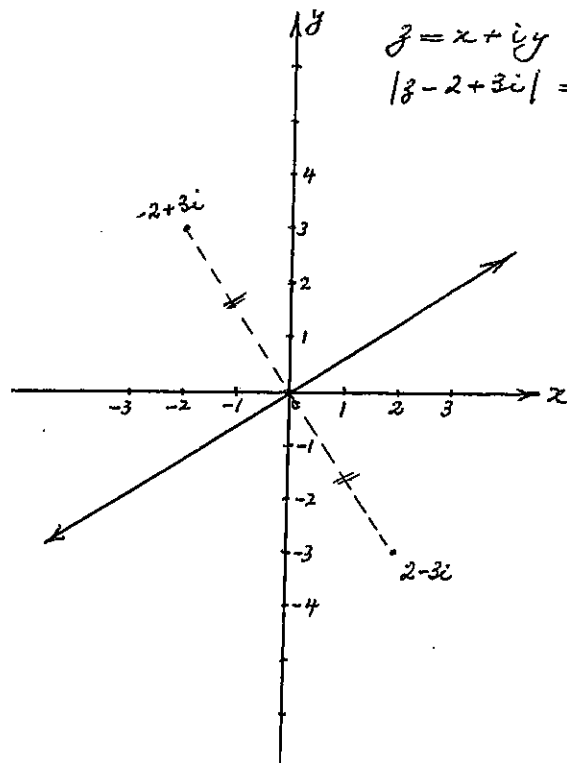
$$= \frac{1}{32} + \frac{1}{32}i$$

$$\therefore a = b = \frac{1}{32}$$

$$(b) \quad (i) \quad |z - 2 + 3i| = |z + 2 - 3i|$$

$$\Rightarrow |z - (2 - 3i)| = |z - (-2 + 3i)|$$

is all points which are equidistant from $2 - 3i$ and $-2 + 3i$

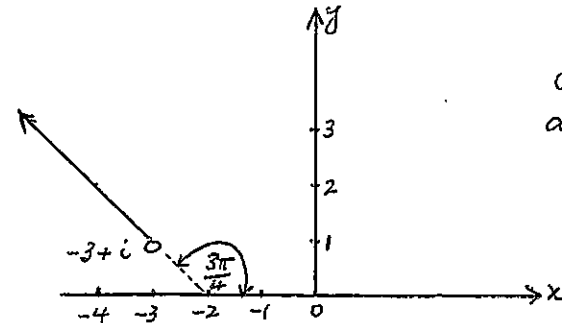


$$z = x + iy \text{ where}$$

$$|z - 2 + 3i| = |z + 2 - 3i|$$

$$(ii) \quad \arg(z + 3 - i) = \frac{3\pi}{4}$$

$$\Rightarrow \arg[z - (-3 + i)] = \frac{3\pi}{4}$$

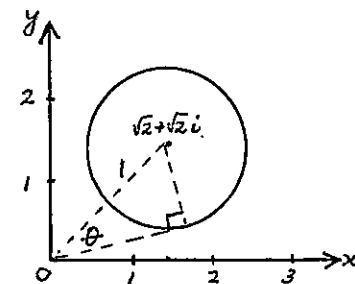


$$z = x + iy \text{ where}$$

$$\arg(z + 3 - i) = \frac{3\pi}{4}$$

(c)

(i)



$$z = x + iy \text{ where}$$

$$|z - \sqrt{2} - \sqrt{2}i| = 1$$

$$\Rightarrow |z - (\sqrt{2} + \sqrt{2}i)| = 1$$

Circle centred at $\sqrt{2} + \sqrt{2}i$ with radius 1.

(ii) See dotted lines in (i) above

$$|\sqrt{2} + \sqrt{2}i| = 2$$

Hence minimum value of $|z|$ is $2 - 1 = 1$

Then the minimum value of $\arg z$ is

$$\arg(\sqrt{2} + \sqrt{2}i) - \theta \text{ where } \sin \theta = \frac{1}{2}$$

$$\Rightarrow \frac{\pi}{4} - \frac{\pi}{6}$$

$$\theta = \frac{\pi}{6}$$

$$= \frac{\pi}{12}$$

$$(d) \quad T \equiv 1$$

$$P \equiv \sqrt{3} + i$$

$$Q \equiv 2 + 2i$$

By similar triangles

$$\frac{|OR|}{|OP|} = \frac{|OQ|}{|OT|}$$

$$\therefore |OR| = \frac{|OP| \cdot |OQ|}{|OT|}$$

$$= \frac{2 \cdot 2\sqrt{2}}{1}$$

$$= 4\sqrt{2}$$

$$\text{and } \arg \vec{OR} = \arg OQ + \theta$$

$$= \frac{\pi}{4} + \arg OP$$

$$= \frac{\pi}{4} + \frac{\pi}{6}$$

$$= \frac{5\pi}{12}$$

$$\therefore R \equiv 4\sqrt{2} \operatorname{cis} \frac{5\pi}{12}$$

QUESTION 14:

(a) Since α, β, γ satisfy equation

$$\alpha^3 - 6\alpha^2 + 3\alpha - 2 = 0 \quad \dots (i)$$

$$\beta^3 - 6\beta^2 + 3\beta - 2 = 0 \quad \dots (ii)$$

$$\gamma^3 - 6\gamma^2 + 3\gamma - 2 = 0 \quad \dots (iii)$$

$$\text{Sum (i), (ii), (iii)} \Rightarrow \alpha^3 + \beta^3 + \gamma^3 - 6(\alpha^2 + \beta^2 + \gamma^2) + 3(\alpha + \beta + \gamma) - 6 = 0$$

$$\text{Now } \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= 6^2 - 2 \times 3$$

$$= 30$$

$$\text{So } \alpha^3 + \beta^3 + \gamma^3 - 6 \times 30 + 3 \times 6 - 6 = 0$$

$$\alpha^3 + \beta^3 + \gamma^3 = 168$$

(b) Let α be zero of multiplicity m
then $P(x) = (x - \alpha)^m Q(x)$ [α not a zero of $Q(x)$]

$$\text{Differentiate } P'(x) = m(x - \alpha)^{m-1} Q(x) + Q'(x)(x - \alpha)^m$$

$$= (x - \alpha)^{m-1} [m Q(x) + Q'(x)(x - \alpha)]$$

$\therefore \alpha$ is a zero of multiplicity $(m-1)$ of $P'(x)$

(c) Since coefficients integers then if z is a zero so is $\bar{z}$.

$$P(x) = [x - (-2-i)][x - (-2+i)](ax^2 + bx + c)$$

$$a, b, c \text{ real} \quad = [(x+2)+i][(x+2)-i](ax^2 + bx + c)$$

$$= [(x+2)^2 - i^2](ax^2 + bx + c)$$

$$= (x^2 + 4x + 5)(ax^2 + bx + c)$$

$$\text{Since } P(x) \text{ is monic, } a=1$$

$$= (x^2 + 4x + 5)(x^2 + bx + c)$$

$$\cdot \text{ constant 5 gives } c=1$$

$$= (x^2 + 4x + 5)(x^2 + bx + 1)$$

$$\cdot \text{ by observation } b=2$$

$$P(x) = (x^2 + 4x + 5)(x^2 + 2x + 1)$$

$$= (x^2 + 4x + 5)(x+1)^2$$

$$\text{Zeros } -2-i, -2+i, -1, -1$$

$$(d) (i) \text{ Let } z = \cos \theta + i \sin \theta$$

$$\text{then } z^3 = (\cos \theta + i \sin \theta)^3$$

$$\cdot \text{ by De Moivre's theorem } z^3 = \cos 3\theta + i \sin 3\theta \quad (I)$$

$$\cdot \text{ on expansion } z^3 = \cos^3 \theta + 3\cos^2 \theta \cdot i \sin \theta + 3\cos \theta (i \sin \theta)^2 + (i \sin \theta)^3$$

$$= \cos^3 \theta - 3\cos \theta \sin^2 \theta + i [3\cos^2 \theta \sin \theta - \sin^3 \theta]$$

II

$$\text{Equating real parts from (I) and II}$$

$$\cos 3\theta = \cos^3 \theta - 3\cos \theta \sin^2 \theta$$

$$\cos 3\theta = \cos^3 \theta - 3\cos \theta [1 - \cos^2 \theta]$$

$$= 4\cos^3 \theta - 3\cos \theta$$

$$(ii) \text{ Let } \cos 3\theta = \frac{1}{2}, \text{ related acute angle } \frac{\pi}{3}$$

in 1st & 4th quad.

$$\therefore 3\theta = \frac{\pi}{3} + 2n\pi, \quad -\frac{\pi}{3} + 2n\pi$$

$$\text{gives } \theta = \frac{2n\pi}{3} + \frac{\pi}{9} \quad * \left(\frac{\pi}{9} (6n \pm 1) \right)$$

$$(iii) \quad 8x^3 - 6x - 1 = 0$$

$$\text{Equivalent to } 2(4x^3 - 3x) = 1$$

$$4x^3 - 3x = \frac{1}{2}$$

$$\text{Let } \cos \theta = x \text{ then } 4\cos^3 \theta - 3\cos \theta = \frac{1}{2}$$

$$\text{equivalent to } \cos 3\theta = \frac{1}{2}$$

$$\text{So solutions from (ii)}$$

$$n=0; \quad \theta = \pm \frac{\pi}{9} \quad \cos \frac{\pi}{9} \quad [= \cos(-\frac{\pi}{9})]$$

$$n=1; \quad \theta = \frac{5\pi}{9} \text{ and } \frac{7\pi}{9}$$

$$\text{Cubic has three solutions } \cos \frac{\pi}{9}, \cos \frac{5\pi}{9}, \cos \frac{7\pi}{9}$$

$$x_1 = \cos \frac{\pi}{9}$$

$$x_2 = \cos \frac{5\pi}{9}$$

$$x_3 = \cos \frac{7\pi}{9}$$

QUESTION 15:

(iv) If $\alpha = \cos \frac{\pi}{4}$ then $\frac{1}{\alpha^2} = \sec^2 \frac{\pi}{4}$

Let $\alpha = a$ then $\frac{1}{\alpha^2} = \frac{1}{a^2} = X$

Require polynomial with X as a zero, $\alpha = \pm \frac{1}{\sqrt{X}}$

Since α is a solution of $8a^3 - 6a - 1 = 0$

then $8\left(\pm \frac{1}{\sqrt{X}}\right)^3 - 6\left(\pm \frac{1}{\sqrt{X}}\right) - 1 = 0$

$[X \neq 0] \quad \frac{8}{X\sqrt{X}} - \frac{6}{\sqrt{X}} - 1 = 0 \quad \left[\frac{-8}{X\sqrt{X}} + \frac{6}{\sqrt{X}} - 1 = 0 \right]$

$8 - 6X - X\sqrt{X} = 0$

$8 - 6X = X\sqrt{X}$

$[-8 + 6X = X\sqrt{X}]$

So $64 - 96X + 36X^2 = X^3$

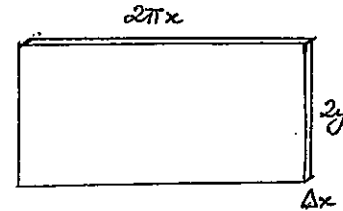
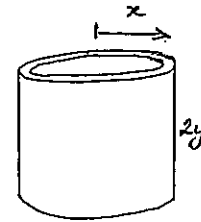
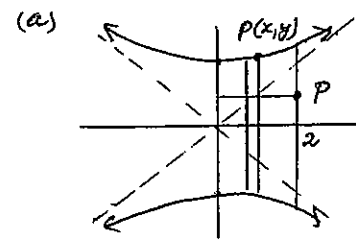
$[64 - 96X + 36X^2 = X^3]$

Require polynomial

$X^3 - 36X^2 + 96X - 64 = 0$

(v) Sum of roots of polynomial $= \frac{b}{a}$

$\therefore \sec^2 \frac{\pi}{4} + \sec^2 \frac{5\pi}{4} + \sec^2 \frac{7\pi}{4} = 36$



Volume of shell is $\Delta V = 2\pi x \cdot 2y \Delta x$
 $= 4\pi xy \Delta x$ ——— ①

where $\frac{y^2}{9} - \frac{x^2}{4} = 1$

ie $\frac{y^2}{9} = 1 + \frac{x^2}{4}$

$\therefore y^2 = \frac{9}{4}(4 + x^2)$

$\therefore y = \frac{3}{2}\sqrt{4 + x^2}$

Then ① $\Rightarrow \Delta V = 4\pi x \cdot \frac{3}{2} \sqrt{4 + x^2} \Delta x$
 $= 6\pi x \sqrt{4 + x^2} \Delta x$

Then the volume of the solid is

$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^2 6\pi x \sqrt{4 + x^2} \Delta x$

$= 6\pi \int_0^2 x \sqrt{4 + x^2} dx$

let $u = 4 + x^2$
 $du = 2x dx$

$= 3\pi \int_0^2 2x \sqrt{4 + x^2} dx$

$= 3\pi \int_4^8 \sqrt{u} du$

$= 3\pi \cdot \frac{2}{3} [\sqrt{u^3}]_4^8$

$= 2\pi [16\sqrt{2} - 8]$

$= 16\pi (2\sqrt{2} - 1) \text{ units}^3$

$$(b) (i) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\Rightarrow \frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$y = \frac{b}{a} \sqrt{a^2 - x^2}$$

$$\text{or } y = \frac{b}{a} \sqrt{a^2 - x^2}$$

Then the area enclosed

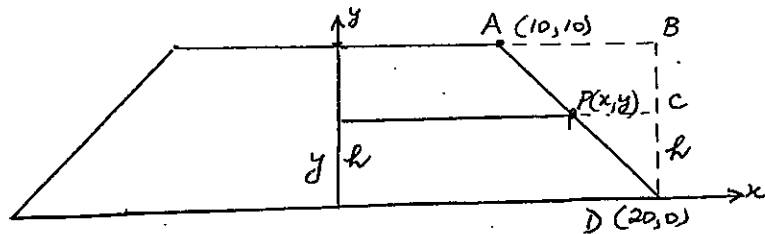
$$= 4 \int_0^a \frac{b}{a} \cdot \sqrt{a^2 - x^2} dx$$

$$= \frac{4b}{a} \cdot \underbrace{\int_0^a \sqrt{a^2 - x^2} dx}_{\text{quadrant of a circle radius 'a'}}$$

$$= \frac{4b}{a} \cdot \frac{1}{4} \cdot \pi a^2$$

$$= \pi ab$$

(ii) (a) Front view



$\triangle ABD \parallel \triangle PCD$ (equiangular)

$$\therefore \frac{AB}{PC} = \frac{BD}{CD}$$

$$\therefore \frac{10}{20-x} = \frac{10}{h}$$

$$\therefore 20-x = h$$

$$\therefore x = 20-h$$

The area of the ellipse at height h is

$$A = \pi ab \quad \text{from (i)}$$

$$= \pi x \cdot \frac{x}{2}$$

$$= \frac{\pi x^2}{2}$$

$\therefore$ Volume of slice is

$$\Delta V = \frac{\pi x^2}{2} \Delta h$$

$$= \frac{\pi}{2} (20-h)^2 \Delta h$$

$\therefore$ Volume of solid is

$$V = \lim_{\Delta h \rightarrow 0} \sum_{h=0}^{10} \frac{\pi}{2} (20-h)^2 \Delta h$$

$$= \frac{\pi}{2} \int_0^{10} (20-h)^2 dh$$

$$= \frac{\pi}{2} \left[\frac{(20-h)^3}{-3} \right]_0^{10}$$

$$= \frac{\pi}{2} \left[\frac{10^3}{-3} - \frac{20^3}{-3} \right]$$

$$= \frac{\pi}{2} \left[\frac{20^3}{3} - \frac{10^3}{3} \right]$$

$$= \frac{3500\pi}{3} \text{ units}^3$$

(c) (i) $xy = 4$
 $= c^2$ where $c = 2$
 $= \frac{1}{2} a^2$

$\therefore a^2 = 8$
 $a = 2\sqrt{2}$ ($a > 0$)

$\therefore$ foci are at (a, a) and $(-a, -a)$
 i.e. $(2\sqrt{2}, 2\sqrt{2})$ and $(-2\sqrt{2}, -2\sqrt{2})$

(ii) (d) normal at P: $y = t^2 x + \frac{2}{t} - 8$

cuts $y = x$ when

$$x = t^2 x + \frac{2-8t}{t}$$

$$x(t^2 - 1) = \frac{8t - 2}{t}$$

$$\therefore x = \frac{8t - 2}{t(t^2 - 1)}$$

$$\therefore N \equiv \left(\frac{8t - 2}{t(t^2 - 1)}, \frac{8t - 2}{t(t^2 - 1)} \right)$$

(b) Gradient of OP: $m_1 = \frac{2}{26}$
 $= \frac{1}{t^2}$

Gradient of PN: $m_2 = t^2$
 (normal at P)

Let $\hat{PON} = \angle$ (angle between $y=x$ and OP) Let $\hat{PNO} = \theta$ (angle between $y=x$ and PN)
 Then $\tan \angle = \left| \frac{1 - \frac{1}{t^2}}{1 + \frac{1}{t^2}} \right|$ Then $\tan \theta = \left| \frac{1 - t^2}{1 + t^2} \right|$

$$= \left| \frac{t^2 - 1}{t^2 + 1} \right|$$

$$= \tan \theta$$

Hence $\theta = \angle$

Then $\triangle PON$ is isosceles.

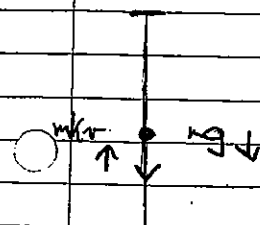
QUESTION 16:

(a) (i) Equation of motion: $F = mg - mkv$

[k positive constant of proportionality]

$$\therefore m\ddot{x} = m(g - kv)$$

$$\ddot{x} = g - kv$$



then $\frac{dv}{dt} = g - kv$

$$\text{so } \frac{dt}{dv} = \frac{1}{g - kv}$$

integrate with respect to v $t = -\frac{1}{k} \int \frac{-k}{g - kv} dv$

$$t = -\frac{1}{k} \ln(g - kv) + c$$

when $t = 0$, $v = v_0$ $\therefore 0 = -\frac{1}{k} \ln(g - kv_0) + c$

$$c = \frac{1}{k} \ln(g - kv_0)$$

$$\text{So } t = -\frac{1}{k} [\ln(g - kv) - \ln(g - kv_0)]$$

$$= -\frac{1}{k} \ln \left[\frac{g - kv}{g - kv_0} \right]$$

gives $-kt = \ln \left[\frac{g - kv}{g - kv_0} \right]$

$$e^{-kt} = \frac{g - kv}{g - kv_0}$$

$$e^{-kt} \left(\frac{g}{k} - v_0 \right) = \frac{g}{k} - v$$

$$\text{So } v = \frac{g}{k} - \left(\frac{g}{k} - v_0 \right) e^{-kt}$$

as required.

(i) When $\ddot{x} = 0$, $g - kv = 0$
 $v = \frac{g}{k}$ terminal velocity

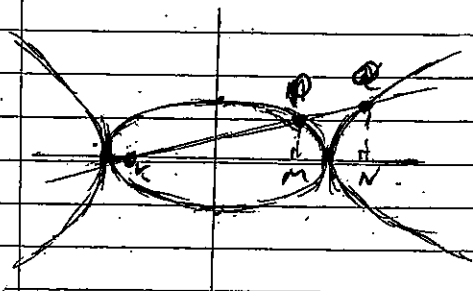
or As $t \rightarrow \infty$, $e^{-kt} \rightarrow 0$

$$\therefore v = \frac{g}{k} - \left(\frac{g}{k} - v_0 \right) \times 0$$

i.e. terminal velocity $v = \frac{g}{k}$

(b) (i) $M(a \cos \theta, 0)$ and $N(a \sec \theta, 0)$

Since $\triangle KPM \parallel \triangle KQN$ (equiangular)



Corresponding sides in proportion

$$\frac{KM}{KN} = \frac{MP}{NQ}$$

$$= \frac{b \sin \theta}{b \tan \theta}$$

$$= \cos \theta$$

(ii) Let distance $OK = d$

then $KM = d + a \cos \theta$

$$KN = d + a \sec \theta$$

So $\frac{KM}{KN} = \frac{d + a \cos \theta}{d + a \sec \theta} = \cos \theta$

$$d + a \cos \theta = d \cos \theta + a$$

$$d(1 - \cos \theta) = a(1 - \cos \theta)$$

$$\underline{d = a}$$

This gives $K(-a, 0)$

(iii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

then $\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$

So $\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$

at $P(a \cos \theta, b \sin \theta)$, gradient of tangent

$$m = -\frac{b^2 (a \cos \theta)}{a^2 (b \sin \theta)}$$

$$m = -\frac{b \cos \theta}{a \sin \theta}$$

Equation of tangent

$$y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$y \sin \theta - b \sin^2 \theta = -\frac{b \cos \theta}{a} x + b \cos^2 \theta$$

$$\frac{b \cos \theta}{a} x + y \sin \theta = b (\sin^2 \theta + \cos^2 \theta)$$

$$x \frac{\cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

Substitute $N(a \sec \theta, 0) \Rightarrow \frac{a \sec \theta \cos \theta}{a} + 0$

$$= 1$$

Tangent passes through N .

(iv) Substitute $M(a \cos \theta, 0)$

Then $\frac{a \cos \theta \sec \theta}{a} - \frac{0 \tan \theta}{b} = 1$

$$1 - 0 = 1$$

M lies on tangent.

Solving $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$ (I)

$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$ (II)

(I) $\times \tan \theta$ $\frac{x \cos \theta \tan \theta}{a} + \frac{y \sin \theta \tan \theta}{b} = \tan \theta$ +

(II) $\times \sin \theta$ $\frac{x \sin \theta \sec \theta}{a} - \frac{y \sin \theta \tan \theta}{b} = \sin \theta$

$$\frac{x}{a} [\sin \theta + \tan \theta] = \tan \theta + \sin \theta$$

$$\therefore \frac{x}{a} = 1$$

$$x = a$$

(lies A and T on line $x = a$, vertical $\therefore \perp$ to x -axis).

(c) Sum $\sum_{r=1}^n r^3 = 1 + 8 + 27 + \dots$

Let $n=1$ LHS $1^3 = 1$

RHS $1^2(1+1)^2 = 4$

$$LHS < RHS$$

true for $n=1$

Let Proposition be true for $n=k$,
 k a positive integer

then $1 + 8 + 27 + \dots + k^3 < k^2(k+1)^2$

For next $n=k+1$

$$1 + 8 + 27 + \dots + k^3 + (k+1)^3$$

$$< k^2(k+1)^2 + (k+1)^3 \quad [\text{from above}]$$

$$= (k+1)^2 [k^2 + k+1]$$

$$< (k+1)^2 [k^2 + k + 1 + 2k + 3]$$

Since $(k+1)^2(3k+5)$ has $(k+1)^2 > 0$
and $3(k+1) > 0$

$$= (k+1)^2 (k^2 + 4k + 4)$$

$$= (k+1)^2 (k+2)^2$$

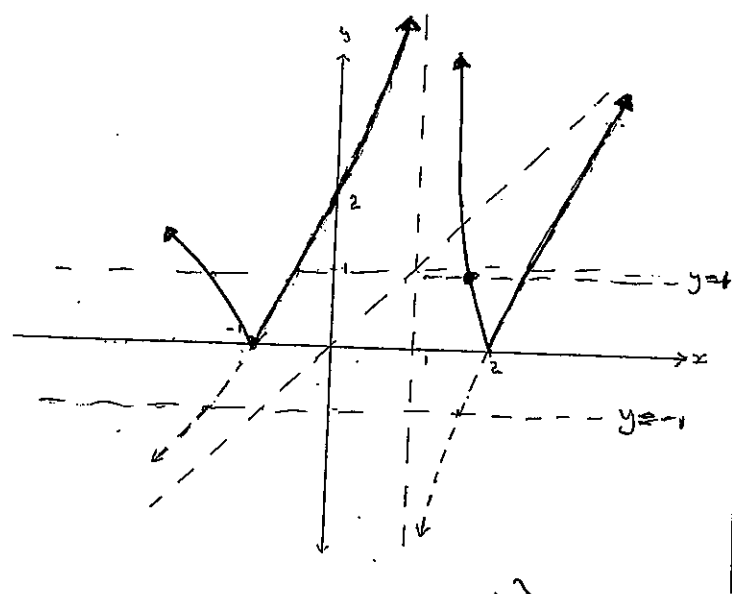
$$= (k+1)^2 [(k+1)+1]^2 \quad \text{as required.}$$

So, if true for $n=k$ we have shown it
is true for next $n=k+1$.

Since true for $n=1$ then it is true for
 $n=2$ and by induction, true for all n .

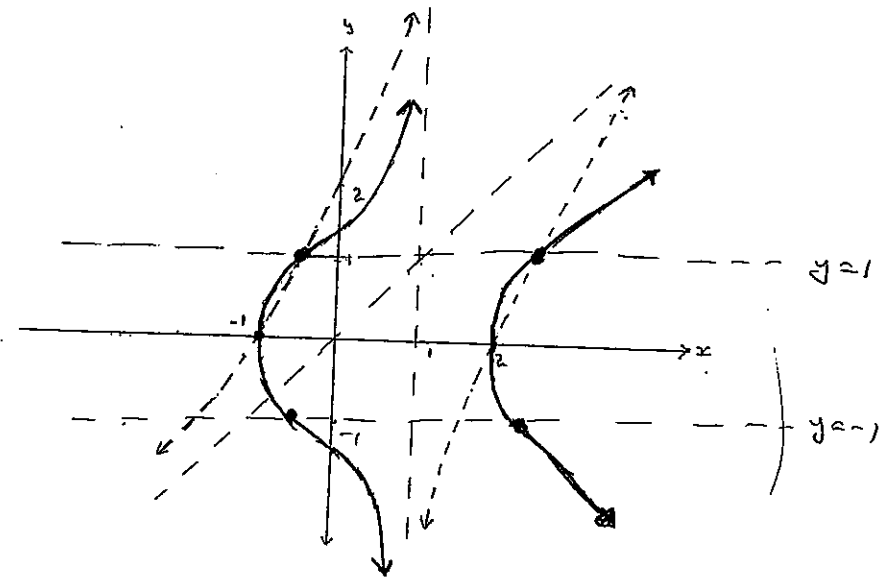
Template Question 12

(α)



Template Question 12

(γ)



Template Question 12

(β)

